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D020711(020)

B. Tech. (Seventh Semester) Examination, April-May 2023

(New Scheme)

(Civil Engg. Branch)

STRUCTURAL ENGINEERING DESIGN-III

Time Allowed : Four hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *Attempt all questions. Part (a) is compulsory and carries 4 marks. Attempt any* ***one*** *from part (b) and (c) from each unit. Assume suitable data, if found missing, and mention it clearly. Draw neat sketches, where necessary. Use of IS 800:2007 and Steel Tables is permitted. All designs are to be done as per the Limit State Method as specified in IS 800:2007.*

Unit-I

1. (a) (I) In the design of framed connections, the bolts connecting the web of the beam with connecting

angles are subjected to :

(i) Single shearing and bearing on the web

(ii) Double shearing and bearing on the web

(iii) Double shearing and not bearing on the web

(iv) No shearing but only bearing on the web

(II) The beam column flexible connections are expected to resist and transfer :

(i) Only shear

(ii) Only moment

(iii) Both shear and moment

(iv) 50% of shear and moment

(b) Design a stiffened seat connection for an ISMB 350 @ 514 N/m transmitting and end reaction of 320 kN (due to factored loads) to column section ISHB 300 @ 576·8 N/m. The steel is of grade Fe410 and bolts of grade 4·6.

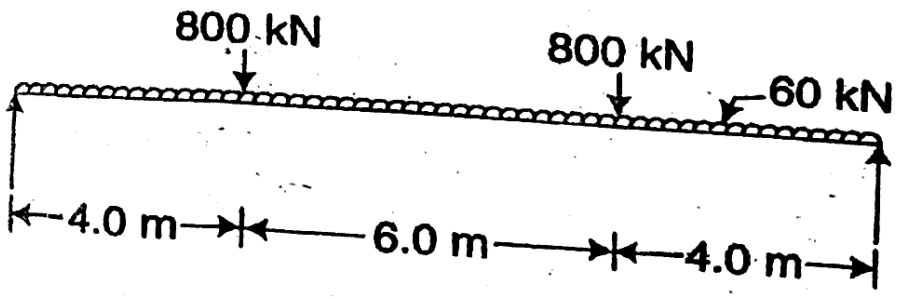
(c) In a framed connection, an ISLB 350 @ 485.6 N/m transmits and end reaction of 220 kN and a moment of 22 kNm, under factored loads to a column ISHB 300 @ 576.8 N/m. Design the connections.

Unit-II

2. (a) Why are end posts required in plate girders?

(b) Design a welded plate girder of 24 m span and laterally restrained throughout. It has to support a uniform load of 100 kN/m throughout the span exclusive of self weight. Design the girder without intermediate transverse stiffeners. The steel for the flange and web plates is of grade Fe 410. Design the cross-section and the end load bearing stiffeners and connections. Yield stress of steel may be asumed to be 250 MPa irrespective of the thickness of the plate used.

(c) Design a plate girder spanning 14 m and loaded as shown in the figure. the self-weight of the plate girder may be assumed to be 60 kN distributed uniformly over the entire span. Use steel of grade Fe410.



Unit-III

3. (a) (I) The surge loads in the gantry girders are :

(i) The gravity loads

(ii) The lateral loads

(iii) The longitudinal loads

(iv) All of the above

(II) Gantry girders are usually designed :

(i) As laterally supported beams

(ii) As laterally unsupported beams

(iii) For a combination of vertical loads and either of the lateral and longitudinal force

(iv) Both (i) and (ii)

(b) A column ISHB 350 @ 661·2 N/m carries an axial compressive factored load of 1700 kN. Design a suitable bolted gusset base. The base rest on M15 grade of concrete pedestal. Use 24 mm diameter bolts of grade 4.6 or making the connections.

(c) Design a hand operarted travelling crane simply supported by grantry girder for the given data :

Span of grantry girder = 8 m

Span of ~~grantry~~ crane girder = 16 m

Crane capacity = 200 kN

Self-weight of crane girder excluding trolley = 200 kN

Self-weight of trolley = 40 kN

Minimum hook approach = 1·20 m

Distance between wheels = 3·5 m c/c

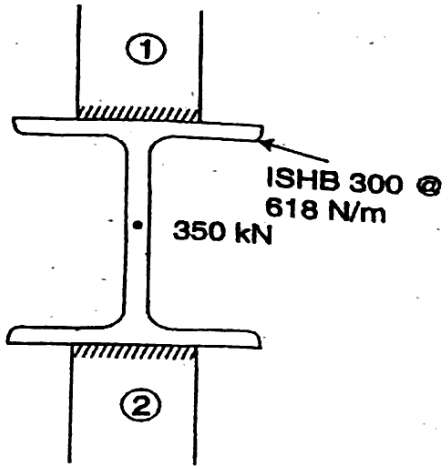
Self-weight of rails = 0.3 kN/m

Unit-IV

4. (a) What is the need of equivalent uniform moment factor in the design of beam-columns?

(b) A column ISHB 300 @ 618 N/m in a framed building supports spandrel beams, as shown in figure below, at its top end. The beams are weled to the column flanges and transfer end reactions of 225 kN(beam 1) and 375 kN (beam 2) and an axial comparessive force of 675 kN from the top storey. The bottom end of the column has a similar beam-to-column arrangement as well as loading. Check the adequency of the column if its effective length is

3.2 m about both the axes. The beam reactions and the loads have been computed from factored laods.



(c) Design an I-section truss member forthe following data.

Effective length of the member : L = 3.5 m

Factored axial tension : T = 450 kN

Factored moment at the two ends of the member about strong axis :

Mz = 35 kNm and 20 kNm, respectively.

Steel of grade : Fe410.

Unit-V

5. (a) What are secondary stresses in roof trusses?

(b) Design a strut in a roof truss for the following data :

Length of the strut = 2.235 m

Factored compressive force = 50 kN (LL + DL)

Factored tensile force = 17.8 kN (LL + DL)

Grade of steel = Fe10

Grade of bolts = 4.6

Bolt diameter = 20 mm

(c) Design the principal rafter of a fink type roof truss for the following data. Design also its connection using 20 mm diameter bolts.

Design Compressive load = 165 kN (due to D.L. and L.L.)

Design tensile load = 60 kN (due to D.L. and W.L.)

Length of rather panel = 2.235 m

Grade of steel = Fe410

Grade of bolts = 4.6